5	(a)		the (value of the) direct current	M1	
			that dissipates (heat) energy at the same rate (in a resistor)	A1	[2]
			allow 'same power' and 'same heating effect'		
	(b)		$\sqrt{2}I_{rms} = I_0$	B1	[1]
	(c)	(i)	power $\propto I^2$ or $P = I^2R$ or $P = VI$	C1	
			ratio = 2.0 (allow 1 s.f.)	A1	[2]
		(ii)	advantage: e.g. easy to change the voltage	B1	
			disadvantage: e.g. cables require greater insulation		
			.... rectification – with some justification	B1	[2]
	(d)	(i)	3.0 A (allow 1 s.f.)	A1	
		(ii)	3.0 A (allow 1 s.f.)	A1	[2]
			Total		[9]
6			0 - + (-1 for each error)	B2	
			+ + 0 (-1 for each error)	B2	